

Processing HTTP Requests & Response

HTTP Request

- 1) Client (web browser) sends a request to a server.
- 2) Request includes:
 - Method
Get, Post, Put, Delete (CRUD op.)
 - URL (Uniform Resource Locator)
 - Headers (metadata eg. User-Agent)
 - Body (optional eg. data sent with a post request.)
- 3) Server receives and process the request.

HTTP Response

- 1) Server sends a response back to the client.
- 2) Response includes:
 - Status code (eg. 200 ok, 404 not found, 500 internal server error)
 - Header (Metadata eg. Content type, Set-Cookie)

- Body (optional e.g. HTML content, JSON data)

3) Client receive and processes the response.

Processing Steps:

1) Request Parsing:

The server breaks down the request into its components (method, URL, header, body).

2) Authentication & Authorization:

The server verifies the client's identity and checks if they have permission to access the requested resource.

3) Resource Identification:

The server determine which resource to retrieve or modify based on the URL.

4) Request Processing

The server perform any necessary actions (e.g. database queries, calculation).

5) Response Generation:

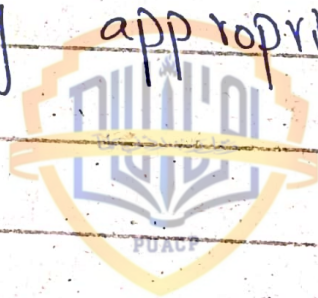
The server creates the response, including the status code, header and body.

6) Response Sending:

The server sends the response back to the client.

7) Response Processing:

The client receives and interprets the response, rendering the content or taking appropriate actions.



Cookies Coordination

Cookies are small text files stored on a user's device by a web browser. They contain information about the user's interactions with a website, such as:

- 1) Preferences (language, font size)
- 2) Login credentials (username, password)
- 3) Session IDs (to track user activity)
- 4) Analytics data (page views, click through rates)

How Cookies work:

1) Cookie Creation:

When a user visits a website, the web browser receives a cookies from web server.

2) Cookie Storage:

The browser stores the cookies on the user's device.

3) Cookie Retrieval:

When the user revisits the website, the browser sends the cookie back to web server.

4) Cookie Processing:

The web server uses the cookie data to personalize the user's experience, authenticate the user or track user behaviour.

Types of Cookies:

1) Session Cookies:

Deleted when the browser is closed.

2) Persistent Cookies:

Remain on the device until they expire or manually deleted.

3) First-Party Cookies:

Set by the website the user is visiting.

4) Third-Party Cookies:

Set by other websites or services (advertising)

Cookie Management:

1) Cookie Settings: user adjust browser setting to accept or reject cookies

2) Cookie Deletion: user can manually delete cookies from device.

3) Cookie Blocking: some browser block cookies by default.

Is cookies are secure?

Cookies security depends on how they are used and implemented.

Cookies are not secure for some reasons:

- 1) Plain text Storage: Cookies are stored in plain text on user's device making them vulnerable and reading by unauthorized parties.
- 2) Unencrypted Data: Cookies often contain unencrypted data, such as user IDs, session IDs which can be accessed by attackers.
- 3) Cookie theft: Cookie can be stolen by using malware or phishing attacks allowing attackers to gain unauthorized access to user data.
- 4) Cookie logging:
Some websites log cookie information which can be a security risk if the logs are not properly secured.

Privacy and P3P

Privacy:-

Privacy refers to the protection of personal information and data collected by website, application and online services.

This include:-

1. Personal information
2. Browsing history and behaviour
3. Search queries & preferences
4. Location data and IP address
5. Cookies and tracking IDs
6. Social media profiles and connection
7. Online transaction and financial data.



P3P

P3P stands for "Platform for Privacy Preferences Project" is a protocol that allows websites to declare their intended use of information they collect about web browser users.

or

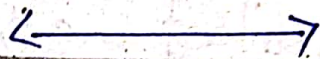
P3P is an emerging industry standard that enables website to express their privacy practices in a standard format that can automatically retrieved and interpreted by user agents.

→ It is developed by W3C (World wide web Consortium).

- Allows websites to declare their privacy policies in a machine-readable format
- Helps user control their personal information when browsing the web.
- Enable browsers to understand and automatically block unwanted cookies

Criticism

P3P faces criticism for being complex, difficult to understand by average human person and lacking enforcement (rules are follows).



Internet Explorer and Microsoft Edge were only major browsers to support P3P.



Complex HTTP Interactions

Complex HTTP interactions typically involve a series of requests and response between a client and a server, often to fulfill tasks such as authentication, authorization, data retrieval and manipulation.

These interaction include various HTTP methods (Get, Post, Put, Delete)

header, status codes and sometimes involve multiple APIs or services communicating with each other.

They often require careful handling of errors, timeouts, rate limits and authentication token to ensure smooth operation and data integrity.

Dynamic Content Delivery:

Dynamic content delivery in web services is a digital strategy that adapts content in real-time based on user interactions, preferences.

→ This is done by using application and scripts that run on the server hosting the website.

It is Useful for:-

- Generates user-specific content
- Scales to handle large user bases and requests.
- Provide security features to protect user data
- Reduce cost by minimizing manual updates and content duplication

Usage

1. Weather Updates: website shows current weather & forecast for user location.
2. Personalized News Feed: on social media platform display data acc. user interaction.
3. Product Recommendations: online website like Amazon shows content based on browsing history.
4. Targeted Advertising:
5. Dynamic Content Blocks: such as trending now or most popular.

Server Configuration

It refers to the process of customizing and setting up a server to meet specific requirements and optimize its performance.

This includes:

- 1) Installing & configuring operating system software and applications.
- 2) Setting up network protocol and connectivity.
- 3) Configuring security settings, such as firewalls and access controls.
- 4) Optimizing server performance, scalability and reliability.
- 5) Configuring backup and recovery systems.

Server Security

Server security involves protecting a server from unauthorized access, use or destruction through.

This includes:

- Network Security (firewalls, intrusion detection)
- Access Control (authentication, permissions)
- Data Security (encryption, backup)
- Software Security (updates, patches)
- Configuration and change management
- Monitoring and incident response
- Physical Security (secure access to server and facility)

This helps ensure the CIA of data and services hosted on the server.

Web Browser Architecture and Processes

Active Browser Pages

An active browser page is a page where the browser performs the logic instead of the server.

- 1) The browser accepts the URL or URI provided by the user and communicates it to the web server, then server retrieves the web page associated with that URL or URI and presents it to the user in the browser window of their client machine.

• Key points

- **Dynamic**: A dynamic page is a page where the server performs the logic and then sends the result to the client.
- **Active**: An active page is a page where the browser performs the logic instead of the server.

Java Script

Java Script is a high-level dynamic and interpreted programming language that is used at front end or client-side scripting on the web.

→ It is use to create logins.

Basic operations:-

- Variable declaration and assignment
- Data type manipulation (no, string, boolean)
- Control structures (if/else, loops)
- Function and function calls
- DOM manipulation (updating webpage content)
- Event handling (responding to user interaction)

1. Variable Declaration:

let x = 5;

const y = "Hello";

2) Data-type Manipulation:

```
let num = 5;  
let str = "Hello";  
let bool = true;
```

3) Control Structures:

```
if (n > 5) {  
  console.log("n is greater than 5")  
}
```

used for output/print data

```
for (let i = 0; i < 5; i++) {  
  console.log(i);  
}
```

4) Functions:

```
function greet(name) {  
  console.log("Hello, ${name}!");  
}
```

user-defined function

```
greet("John");
```

Output : Hello, John!

Document
object
Model

5) DOM Manipulation:

document.getElementById("myId").

JS use to access DOM
innerHTML = "New Text";

selecting element

6) Event Handling:

```
document.getElementById("myButton").addEventListener("click", function () {  
    console.log("Button clicked!");  
});
```

Usage of JS

It is often used to add interactive elements to website, such as:

- Animation and effect
- Form submission and validation
- Dynamic updates to webpages
- Responsive user interface elements
- web application and games.

It is a combination of HTML, CSS and JavaScript to create interactive and dynamic web pages.

Features:

- Tags and their properties can be changed using DHTML.
- It is used for real-time positioning.
- Dynamic fonts can be generated using DHTML.
- It makes a webpage dynamic and used to create animations, games application along with providing new ways of navigating through websites.
- Functionality of webpage is enhanced due to usage of low bandwidth effect by DHTML.
- It also facilitates use of method, event, properties and code.

Usage

- DHTML is used for designing the animated and interactive web pages that are developed in real-time.

- It allows authors for adding the effect on their page
- It allows the page authors for including drop-down menus/rollover button.
- This term is also used to create various browser-based action games.
- DHTML is used to add the trick on various websites, which needs to refresh their content automatically.

AJAX (Asynchronous JavaScript and XML)

It is a set of web development techniques that uses various web technologies on the client-side to create asynchronous web application.

→ It allows webpages to update asynchronously by exchanging data with a web server.

- AJAX is not a programming language. It uses a combination of JS, HTML and DOM (to display/use data) and a browser built in XMLHttpRequest object (to request data from a web server).

- AJAX allows web pages to be updated asynchronously.

By using AJAX it is possible to update parts of webpage without reloading the whole page.

AJAX is used for:

- 1) Reading data from a web server after the page has loaded
- 2) Updating a web page without reloading the page.
- 3) Sending data to web server in the background.

How it works?

- 1) An event occurs in web page (page is loaded, a button is clicked).
- 2) An XMLHttpRequest object is created by JavaScript.
- 3) The XMLHttpRequest object sends a request to a web server,
 - Server processes the request
 - Server sends a response back to the web page
 - Response is read by JS
 - Proper action (like page update) is performed by JS

Drawback for XMLHttpRequest

Modern browser use Fetch API instead of XMLHttpRequest object.

Fetch API

Its interface allows web browser to make HTTP requests to web servers.

Fetch API is used as a part of AJAX application.



JSON (JavaScript Object Notation)

It is a lightweight, easy-to-read and write, data interchange format.

→ It is a text-based format used for exchanging data b/w web servers, web applications and mobile apps.

JSON is based on two structures.

1) Object:

Unordered collection of key-value pairs, represented by curly braces

```
{ }
```

2) Arrays:

Ordered collection of value

represented by square brackets
[].

JSON data is composed of.

- Keys (strings, enclosed in double quotes)
- Values (string, no., boolean, arrays, objects or null)
- Separators (commas, separate key values pair and array element).

Example:

```
{  
  "name": "IMRAN",  
  "Age": "63",  
  "Occupation": "Politician",  
  "address": {  
    "city": "Lahore",  
    "Province": "Punjab",  
    "place": "Adiala Jail",  
  },  
  "interests": ["cricket", "leader"]  
}
```

JSON is used for:

- Data exchange b/w web servers and web applications
- Data storage and retrieval in web storage and database
- configuration files and settings
- API data exchange

JSON is simple, efficient and platform independent data format